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Studierichting: Informatica
Oefeningenreeks 4A nr. 43.7

$$\int \frac{1-x^4}{3x^5+x^4}dx$$

$$\frac{1-x^4}{3x^5+x^4} = \frac{1-x^4}{x^4(3x+1)}$$

$$\frac{1-x^4}{x^4(3x+1)} = \frac{A}{x^4} + \frac{B}{x^3} + \frac{C}{x^2} + \frac{D}{x} + \frac{E}{3x+1}$$

$$A = \left.\frac{1-x^4}{3x+1}\right|_{x=0} = 1$$

$$E = \left.\frac{1-x^4}{x^4}\right|_{x=-\frac{1}{3}} = 80$$

$$\frac{1-x^4}{x^4(3x+1)} - \frac{1}{x^4} = \frac{1-x^4-(3x+1)}{x^4(3x+1)} = \frac{-x^4-3x}{x^4(3x+1)} = \frac{-x(x^3+3)}{x^4(3x+1)} = \frac{-(x^3+3)}{x^3(3x+1)}$$

$$B = \left.\frac{-(x^3+3)}{3x+1}\right|_{x=0} = -3$$

$$\frac{-(x^3+3)}{x^3(3x+1)} + \frac{3}{x^3} = \frac{-(x^3+3)+3(3x+1)}{x^3(3x+1)} = \frac{-x^3-3+9x+3}{x^3(3x+1)} = \frac{-x^3+9x}{x^3(3x+1)} =$$
$$\frac{x(-x^2+9)}{x^3(3x+1)} = \frac{-x^2+9}{x^2(3x+1)}$$

$$C = \left.\frac{-x^2+9}{3x+1}\right|_{x=0} = 9$$

$$\frac{-x^2+9}{x^2(3x+1)} - \frac{9}{x^2} = \frac{-x^2+9-9(3x+1)}{x^2(3x+1)} = \frac{-x^2+9-27x-9}{x^2(3x+1)} = \frac{-x^2-27x}{x^2(3x+1)} =$$
$$\frac{x(-x-27)}{x^2(3x+1)} = \frac{x+27}{x(3x+1)}$$

$$D = \left.\frac{x+27}{3x+1}\right|_{x=0} = 27$$

$$\int \frac{1-x^4}{3x^5+x^4}dx = \int \left(\frac{1}{x^4} - \frac{3}{x^3} + \frac{9}{x^2} + \frac{27}{x} + \frac{80}{3x+1}\right) dx$$
$$= \frac{-1}{3x^3} + \frac{3}{2x^2} - \frac{9}{x} - 27\ln|x| + \frac{80}{3}\ln|3x+1| + C$$

References